

Comprehensive Benchmark of Hybrid *de novo* Transcriptome Assembly Approaches

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Background:

When high-quality reference genomes for non-model organisms are unattainable due to financial (insufficient funding) or technical (highly repetitive and large genomes), *de novo* transcriptome assemblies provide a valuable alternative as they capture the functional output of the genome and provide a basis for a variety of analysis such as gene-expression analysis or positive selection analysis.^{1,2} Transcriptomes are typically assembled from RNA-Seq data generated either with Illumina, which produces large quantities of highly accurate but highly fragmented reads or with ONT / PacBio which produce much longer reads albeit at lower quantities and typically lower nucleotide-level accuracy. Hybrid approaches attempt to leverage the respective advantages of both types of reads. We have identified and tested three strategies for hybrid *de novo* assembly:

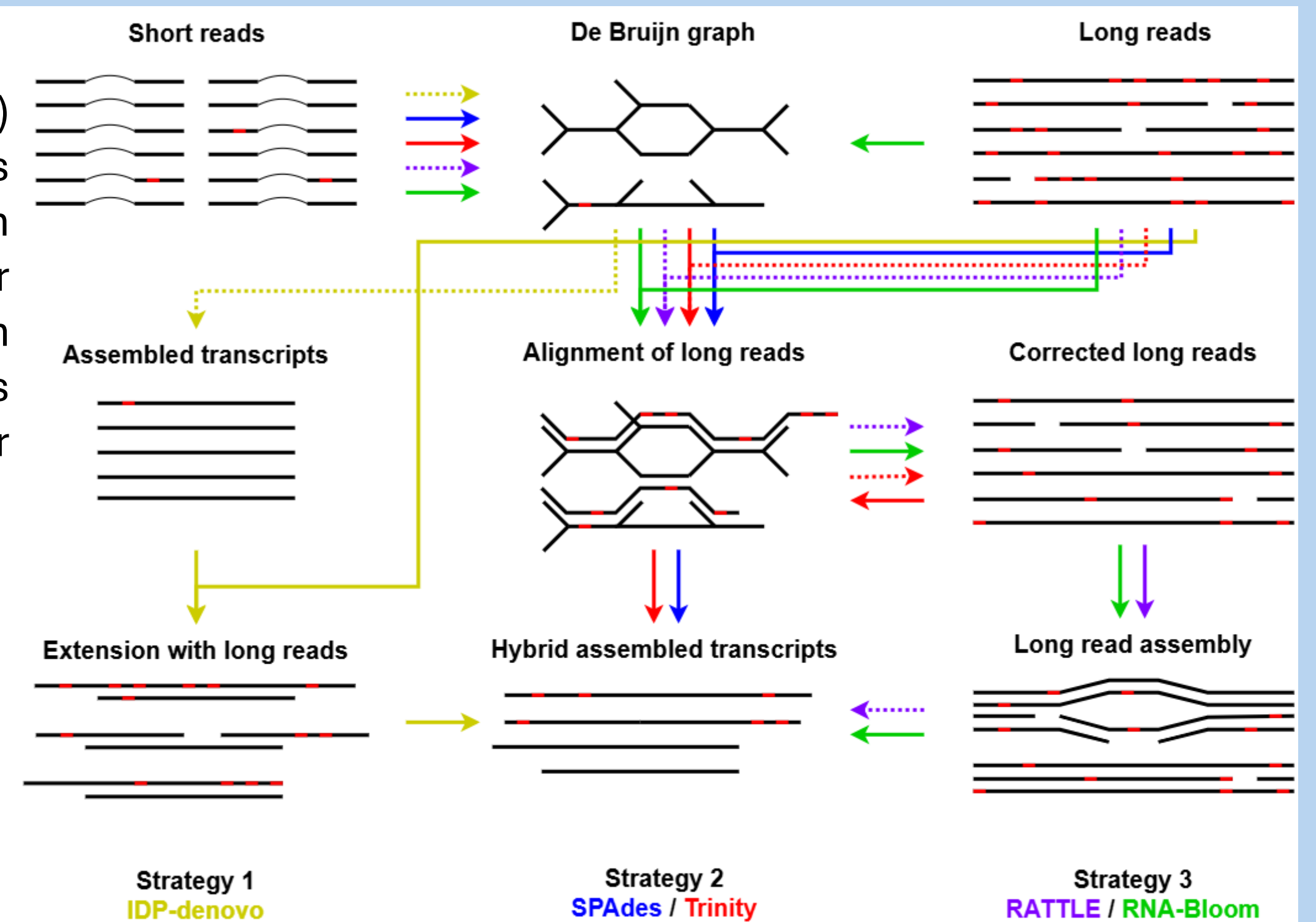
Strategy 1 (IDP-denovo³):

Strategy 2 (SPAdes^{4,5}/Trinity⁶):

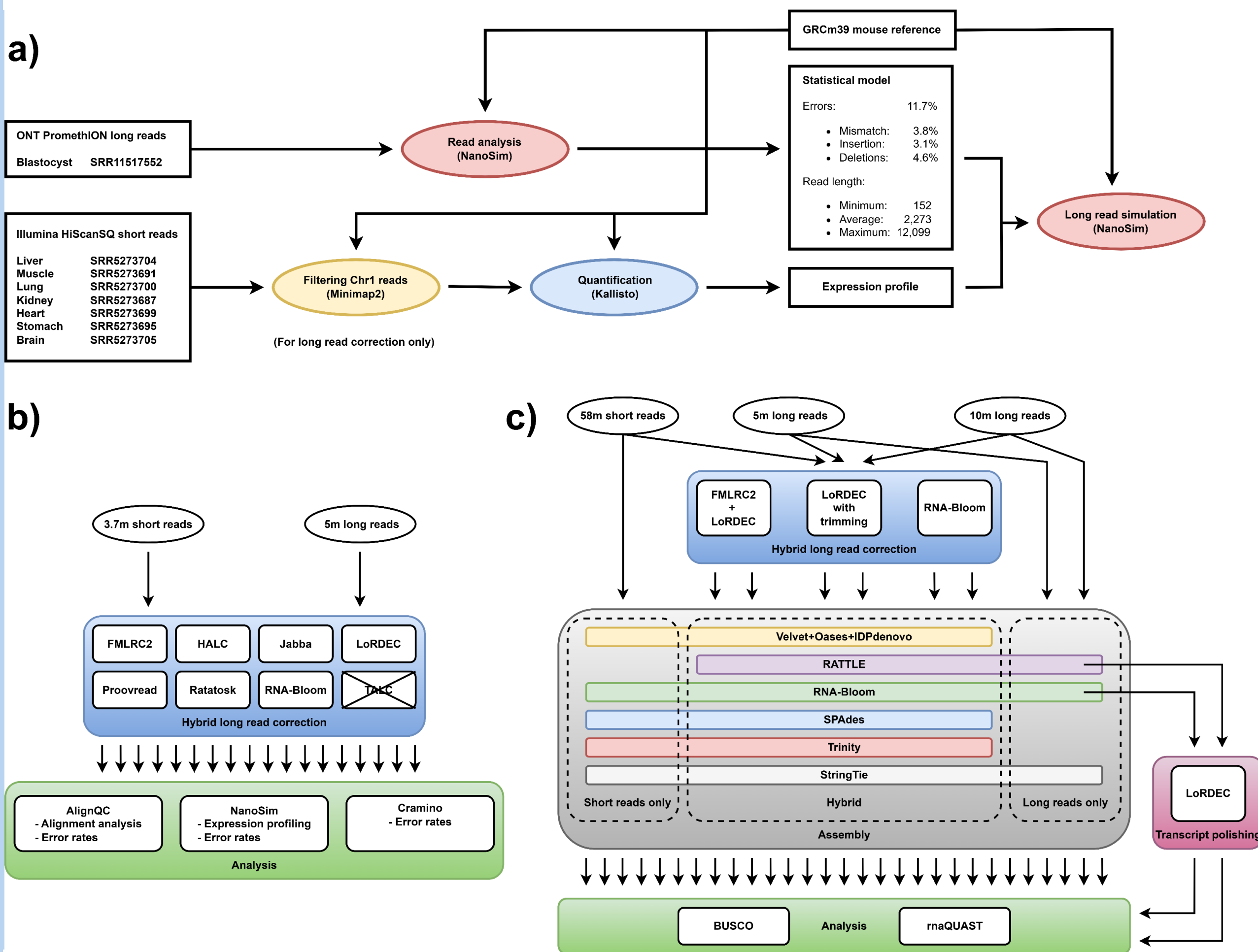
Strategy 3 (RATTLE⁷/RNA-Bloom^{8,9}): Long reads are mapped to the short read k-mer / de Bruijn graph to correct base-calling errors before long read only assembly. Similarly, transcripts can be polished after long read only assembly.

Extension of transcript fragments after short read only assembly

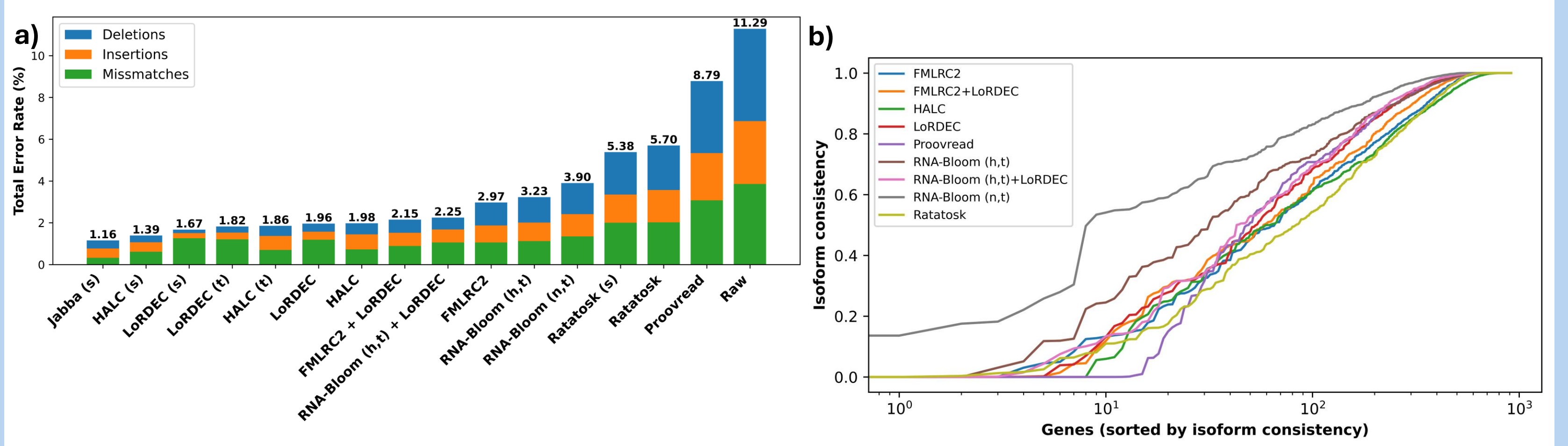
Long reads are mapped to the short read k-mer / de Bruijn graph to identify paths representing transcripts and closing of coverage gaps during assembly



Study design:



(Hybrid) long read correction:



a) Nucleotide-level error rates:

- Correction tools remove base-calling errors at different rates
 - Trimming (t) and splitting (s) further reduces error rates by removing uncorrected regions
 - Jabba achieves lowest error rates but either removes reads entirely or leaves them fragmented
 - RNA-Bloom does not require additional short reads (n) but will remove more errors with them
- b) Gene isoform consistencies (cosine similarity of transcript count vectors before/after correction):
- Long read correction alters expression profiles and exon composition of long reads (See also Lima et al, 2020)
 - The expression-level aware RNA-Bloom better preserves the expression profile, especially when used without additional short reads

c) Combining tools decreases both error rate and mean gene isoform consistency

a) Benchmark dataset creation:

Real Illumina short read pairs from multiple tissues of the mouse¹⁰ and simulated ONT long reads¹¹ with matching expression profile¹².

b) Long read correction benchmark:

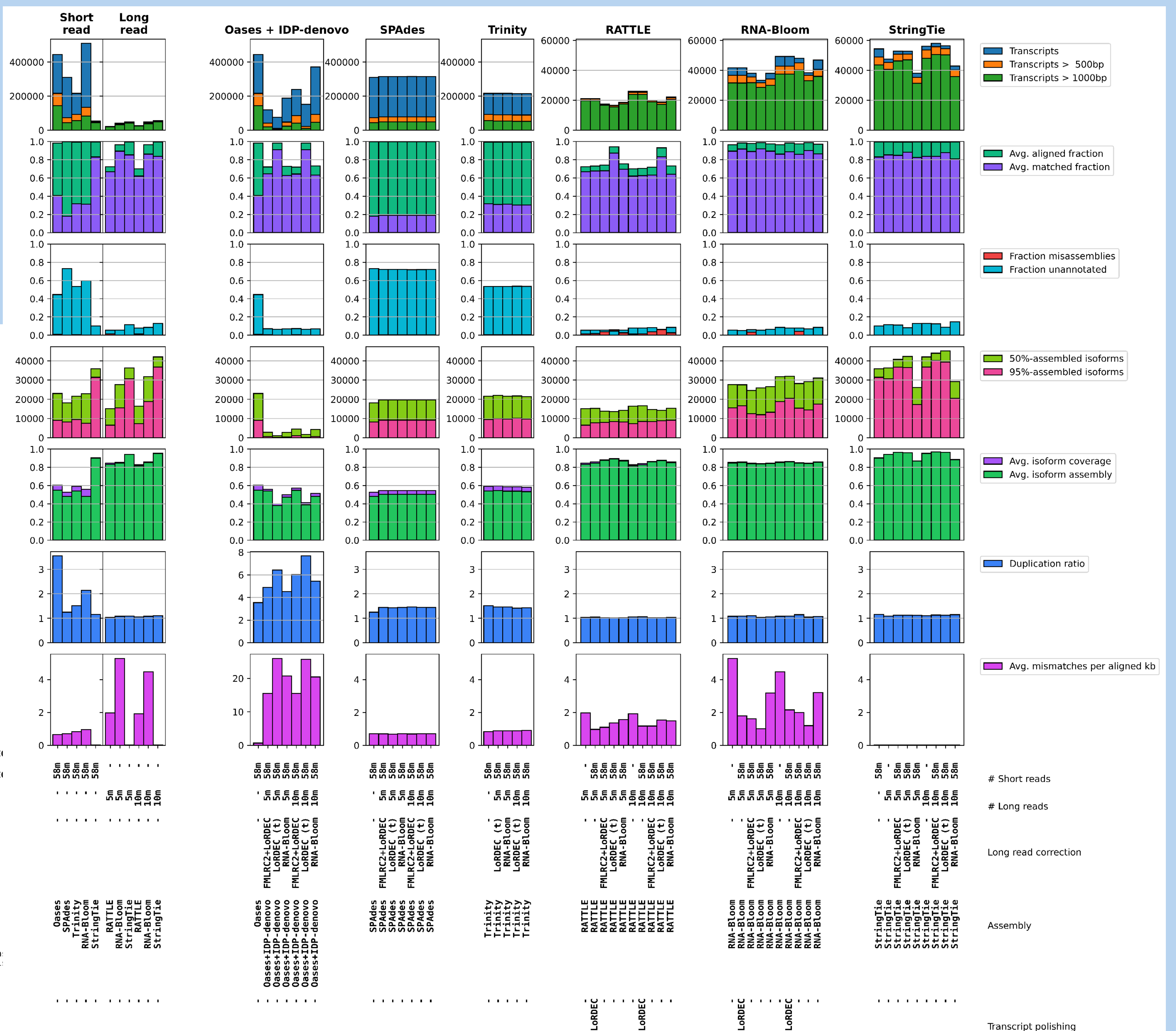
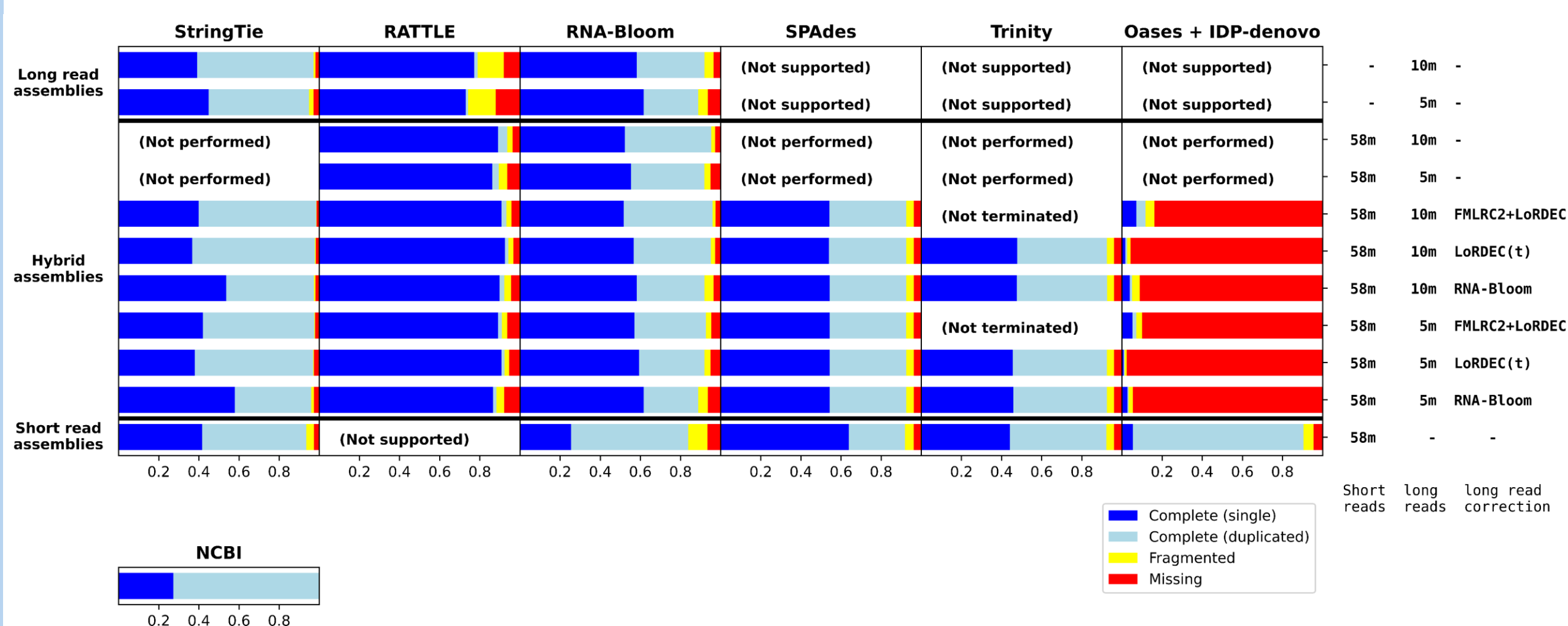
Using a reduced-size data set of 3.7 million short and 1 million long reads we tested^{11, 13, 14} 8 correction tools^{9,15,16,17,18,19,20,21}

c) *De novo* assembly benchmark:

Comparison^{22,23,24} of hybrid *de novo* assemblies both with each other as well as with their non-hybrid counterparts^{3,4,5,6,7,8,9,25}. Additionally, genome-guided assemblies^{26,27,18,29,30,31,32} were generated as a reference.

(Hybrid) *de novo* assembly:

- Short reads: nucleotide-level accuracy vs Long reads: structural accuracy and completeness
- Hybrid IDP-denovo, SPAdes and Trinity assemblies show little/no improvement
- Hybrid correction improves nucleotide-level accuracy in long read transcriptomes and allows detection of additional BUSCOs.
- Isoform structure alterations during long read correction propagate into long read assemblies and compromise both completeness and structural accuracy
- Post-assembly transcript polishing improves nucleotide-level accuracy while maintaining completeness and structural correctness of long read assemblies



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